

# Why Odd Numbers Build Perfect Squares

Interactive Proof Project / July 2026

## THEOREM

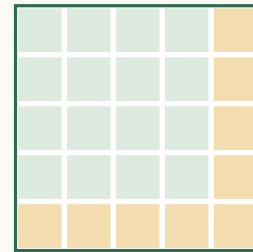
For every natural number  $n$ , the sum of the first  $n$  odd numbers is  $n^2$ :  
 $1 + 3 + 5 + \dots + (2n - 1) = n^2$ .

## PROOF BY INDUCTION

Write  $S_n$  for  $1 + 3 + \dots + (2n - 1)$ . For  $n = 0$ , the sum is empty, so  $S_0 = 0 = 0^2$ . Now assume  $S_n = n^2$ . The next odd number is  $2n + 1$ , hence

$$S_{n+1} = S_n + (2n + 1) = n^2 + 2n + 1 = (n + 1)^2.$$

A 4 x 4 square grows to 5 x 5



The new border has  $4 + 5 = 9$  cells.

Therefore the identity holds for  $n + 1$ , and induction proves it for every natural number  $n$ .

## WHY THE ALGEBRA MATCHES THE PICTURE

To enlarge an  $n$  by  $n$  square, attach one column of  $n$  cells and one row of  $n + 1$  cells. The new L-shaped border contains  $2n + 1$  cells - exactly the next odd number. The equation  $(n + 1)^2 - n^2 = 2n + 1$  is the same growth step written algebraically.

## FORMALIZATION NOTE

Lean defines `oddSum` recursively: `oddSum 0 = 0` and `oddSum (n + 1) = oddSum n + (2 * n + 1)`. The theorem follows the same induction as the paper; simplification expands the successor square and closes the arithmetic step.